Independent reproduction check

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

Check another operator's result against the same recorded computation

Resolve metric and scoring disagreements from artifacts without inventing a successful reproduction.

Keep one habit: Question and contract → complete main ledger → separate picture → boundary probes → investigation → independent attempt.

MAIN LESSON · SECTION 1

Check another operator's result against the same recorded computation

> Resolve metric and scoring disagreements from artifacts without inventing a successful reproduction.

Learning goals

I can explain the **mechanism** and terminology of independent reproduction check.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Retrieve before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Peer review and response. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Writing the methods section. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Robustness across conditions. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Starting explanation

- Recall Peer review and response. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

My starting explanation

End goal

Explain independent reproduction check using the records and boundaries of this fictional investigation.

▼ **Earlier wording, alternatives and picture transcripts**

Your learning target

Remember before reading

This week you will practise:

Explain the **mechanism** and terminology of independent reproduction check

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Retrieve what you already know

This week’s end goal

MAIN LESSON · SECTION 2

Understand the question and evidence contract

> Resolve metric and scoring disagreements from artifacts without inventing a successful reproduction.

Start with the research problem

An independent reproduction check asks whether another operator can recover the stated computation from the recorded materials without hidden help. A disagreement is a clue to inspect, not a score to average away. Compare exact data roles, code, metric definitions, environment and raw outputs. Record any assistance and missing files.

Rerunning saved cases is different from testing a new sample, and this course’s local calculation does not itself create a second independent operator.

Original concept explanation

Independent reproduction checks whether another operator can use the recorded artifacts to recover the stated computation without hidden help.

Complete case inputs, rules and limits

Reproduction item	Complete supplied exercise contract
Main metric conflict	Paper names mean squared error; supplied script averages absolute errors

Reproduction item	Complete supplied exercise contract
Main numerical values	Residuals and reported scores not supplied
Added distinct fixture	Predictions 1, 3; targets 1, 1; residuals 0, 2
Separate picture	Original score 5/6; rerun 4/6; only Q3 scoring differs
Picture generation	No new model outputs generated
Changed archive	Q3 saved file missing
Independent case	Different validation split used because split IDs absent
Evidence limit	No full-study independent reproduction or new-data replication performed by the demonstration

Name given and missing evidence

List the given facts and one fact that is not supplied.

▼ Earlier wording, alternatives and picture transcripts

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Trace the complete main case before interpreting it

> Follow each evidence-to-conclusion step and retain unknown quantities.

Original declared procedure

Provide manifest, fixture data, code, environment and expected metric definition. Record assistance and mismatches; reproduction is not new-data replication.

Complete main-case evidence ledger

Main metric audit	Supported statement
Named metric	MSE averages squared residuals
Implemented metric	MAE averages absolute residuals
Contract mismatch	Different computation; inspect/correct script or report with provenance
Actual original discrepancy size	Unknown; no main residual values supplied
Distinct fixture residuals	0 and 2
Fixture MAE	$(0+2)/2 = 1$
Fixture MSE	$(0^2+2^2)/2 = 2$
Fixture purpose	Demonstrates possible difference, not the missing original run
After correction	Preserve old artifacts/results and record a new documented rerun

State the scope and explain the visual

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ **Earlier wording, alternatives and picture transcripts**

Independent reproduction check — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Operator gets different MSE because the supplied script averages absolute errors while paper names squared error.	Provide manifest, fixture data, code, environment and expected metric definition. Record assistance and mismatches; reproduction is not new-data replication.	Metric contract mismatch explains a candidate discrepancy. Correct/report script and rerun; preserve original artifact versions and results.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Connect the picture and the explanation

MAIN LESSON · SECTION 4

Predict the changed case before reading the explanation

> Keep the rule and evidence boundary visible.

Main task and response

- Operator gets different MSE because the supplied script averages absolute errors while paper names squared error. Show the intermediate steps and give the conclusion with its scope.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Predict the pictured change

- Change: Q3's saved file is missing from the archive.

▼ **Check after predicting**

Prediction: Mark exact reproduction incomplete until the required artefact is recovered or the limitation disclosed.

Reason: A missing record prevents checking the contested decision.

▼ **Earlier wording, alternatives and picture transcripts**

Compare a familiar case with a changed case

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Connect the main result with the separate pictured case

> Keep the two scenarios and their supplied evidence distinct.

Main case and original checked reasoning

Operator gets different MSE because the supplied script averages absolute errors while paper names squared error.

Metric contract mismatch explains a candidate discrepancy. Correct/report script and rerun; preserve original artifact versions and results.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Separate pictured inputs and complete explanation

An independent rerun with a disagreement. A second team follows the saved protocol on the same fixed data and obtains a different score.

Original correct count is 5/6. Rerun is 4/6. Case Q3 is the only scoring disagreement. No new model **output**s are generated.

The separate picture’s original correctness count is five of six, while the supplied rerun is four of six. Only case Q3 has a scoring disagreement; no new model outputs are generated. Inspect Q3’s saved output, reference label and rubric before deciding how it should score. Do not simply take the higher count or average incompatible scoring judgments into a supposedly resolved result.

If Q3’s saved file is missing, exact verification of that case is unresolved; record the gap and any partial check honestly. The existing counts do not reveal the content needed for adjudication, and no program here can restore that missing evidence.

Read the labelled representation

A scoring disagreement needs the record and rubric

Original supplied score

5 / 6

Rerun supplied score

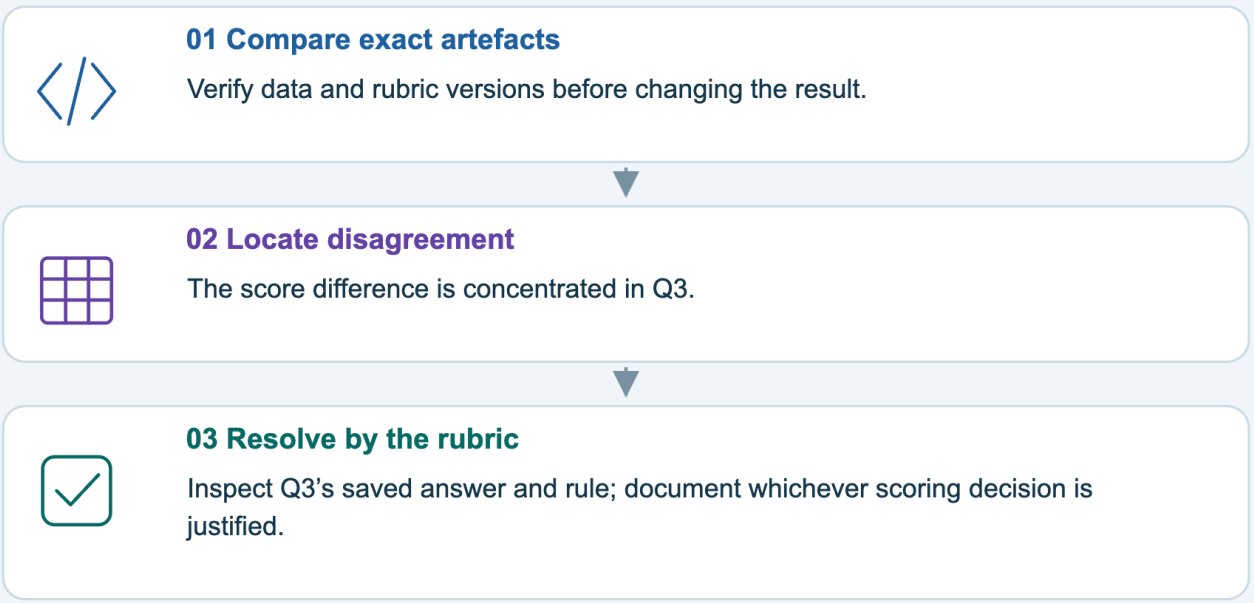
4 / 6

Only disputed scoring case: Q3

Inspect saved output + reference label + scoring rule.
If Q3’s saved file is missing, verification remains unresolved.
No new model outputs are generated; this is not new-data replication.

AI ACADEMY / AI-7/33 · 02 · TRACE THE CASE

Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Compare exact artefacts	Verify data and rubric versions before changing the result.
2. Locate disagreement	The score difference is concentrated in Q3.
3. Resolve by the rubric	Inspect Q3's saved answer and rule; document whichever scoring decision is justified.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

An independent reproduction check can uncover interpretation or implementation differences. Do not average incompatible scores or simply choose the higher one.

Complete picture evidence and limits

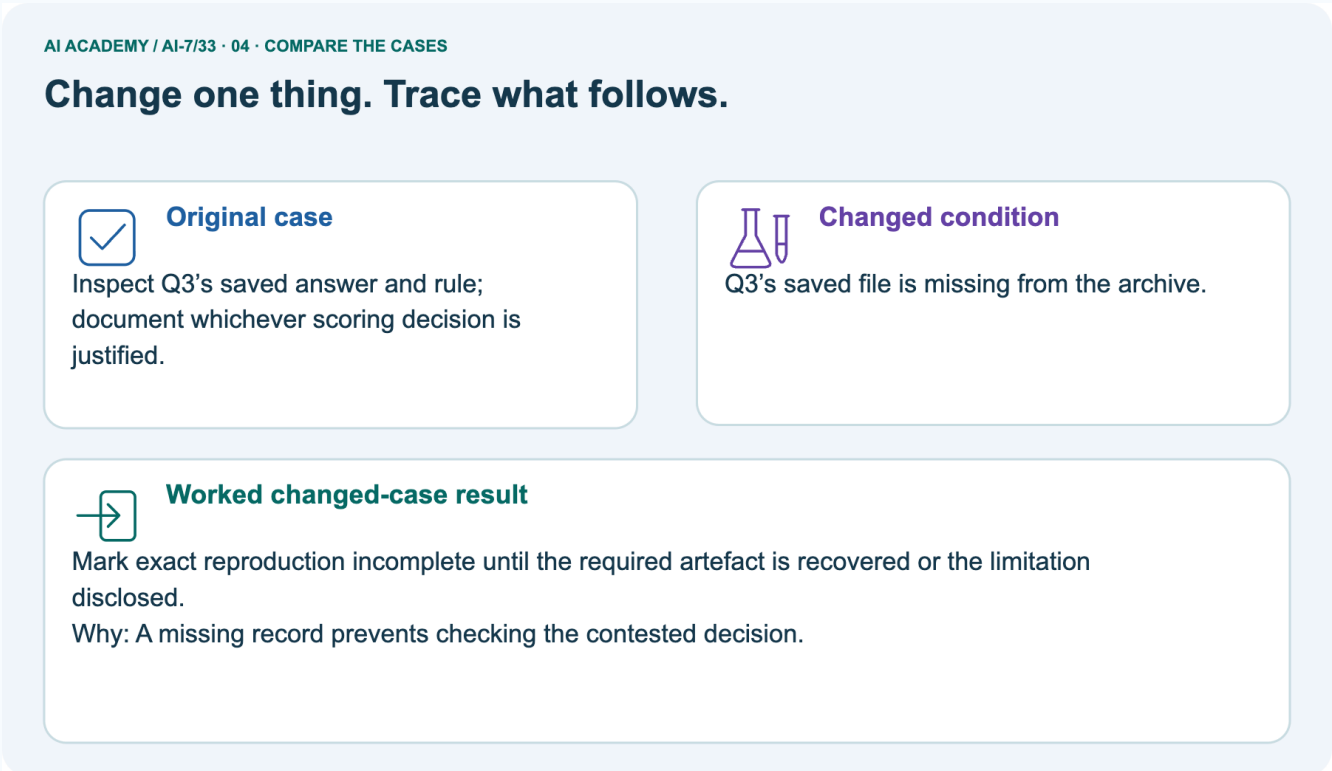
Separate scoring check	Supplied evidence or action
Original correctness	5/6 ≈ 83.333%
Rerun correctness	4/6 ≈ 66.667%
Descriptive difference	One case, about 16.667 percentage points
Disputed case	Q3 only
What to inspect	Saved output, reference label, rubric and scoring implementation
Adjudicated correct score	Not supplied; do not invent a winner
Q3 file missing	Exact verification unresolved; document missing artifact

Separate scoring check	Supplied evidence or action
Data role	Same saved outputs, not new-sample replication

Predict the changed condition

- Change one thing: Q3’s saved file is missing from the archive. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain what remains unknown

- This check concerns saved **output**s and scoring, not new-sample replication or future model behaviour.

- Read the labelled picture: Locate the Q3 scoring disagreement between 5/6 and 4/6. What should the teams inspect before choosing a score, and what limitation follows if the saved Q3 file is missing?

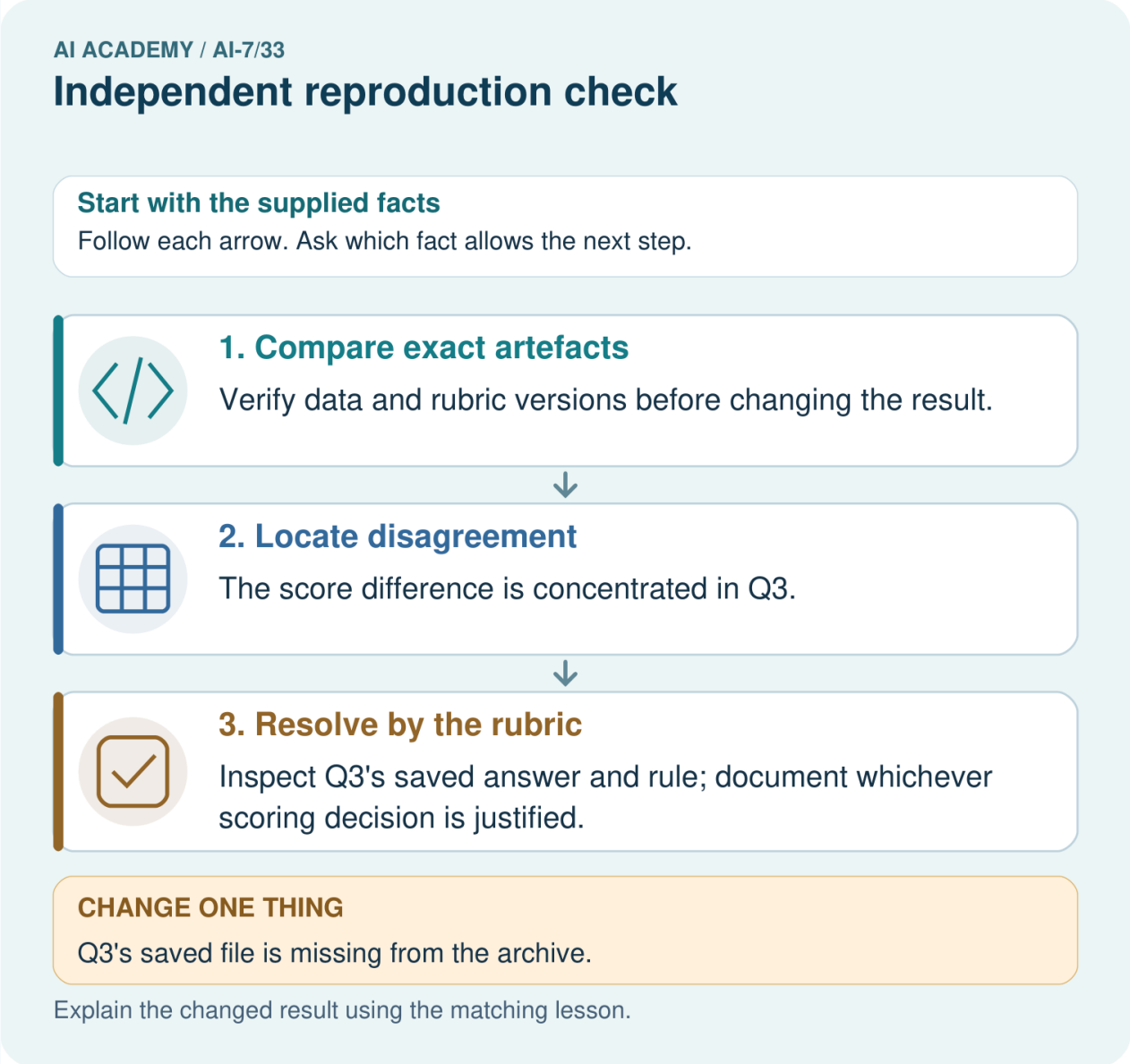
<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.
</details>

A pictorial worked case: Independent reproduction check

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Find the first unsupported inference and repair it

> Use the supplied conditions rather than inventing missing evidence.

Original qualification and repair setup

A successful reproduction repeats the original **evidence** and may preserve its original design limitations.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Complete boundary and misconception probes

Reproduction audit	Checked interpretation
Same metric name, different formula	Compare implementations, not just labels
MAE and MSE differ in a fixture	Demonstrates metric contract; not proof of every original mismatch
Original and rerun scores disagree	Inspect raw records and scoring rule
Choose higher result	Does not resolve incompatible judgments
Missing Q3 file	Mark exact check unresolved; do not fabricate evidence
Different validation split	Not the same evaluation materials
Other operator received help	Record assistance rather than call it unaided
Same-data rerun	Not new-data replication or guaranteed future behavior

Write and justify the repair

- A report omits this qualification: A successful reproduction repeats the original **evidence** and may preserve its original design limitations.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

State one repair and the checks you would rerun.

▼ Earlier wording, alternatives and picture transcripts

Locate the earliest unsupported step

MAIN LESSON · SECTION 8

Practise the mechanism with progressively less help

Keep the selected independent case for a first attempt.

Shared explanation

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

> **Explain your prediction to a partner before comparing results.**

Supported workbook trace

Use workbook W2 for the connected example “Independent reproduction check”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Individual scope check

- Use the worked case for Independent reproduction check. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Picture response

- Locate the Q3 scoring disagreement between 5/6 and 4/6. What should the teams inspect before choosing a score, and what limitation follows if the saved Q3 file is missing?

> **My explanation using a pictured step and the given facts**
>
> | |
> |---|
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> | |
> | |

Reserve the independent task

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

Practise with less help: Trace the worked case

Check your reasoning

Explain the picture

MAIN LESSON · SECTION 9

Investigate, record the evidence and check the limits

> Distinguish a prediction, a paper trace and an actual local execution.

Current investigation

Operator gets different MSE because the supplied script averages absolute errors while paper names squared error. Show the intermediate steps and give the conclusion with its scope.

Complete investigation instructions

1. Audit the named MSE versus implemented MAE contract on paper; the main scenario has no numerical residuals. Then predict the added fixture's MAE one and MSE two before running or tracing the complete code. Keep this fixture separate from the original report. For the scoring picture, list the Q3 artifacts needed to adjudicate 5/6 versus 4/6 and what remains unresolved when the saved file is missing.

   The code performs only a local metric calculation and prints stipulated disagreement metadata. It does not contact another operator, generate model outputs or claim an independent study was reproduced. Reserve the missing-split independent task.

```
"""A distinct numerical fixture demonstrating metric contracts, not a real reproduction."""
from fractions import Fraction

def error_metrics(predictions,targets):
 if not isinstance(predictions,(list,tuple)) or not isinstance(targets,(list,tuple)) or not
predictions or len(predictions)!=len(targets):raise ValueError('same nonempty case lists
required')
 if any(type(v) is not int for v in list(predictions)+list(targets)):raise ValueError('exact
integer fixture values required')
 errors=[p-y for p,y in zip(predictions,targets)];n=len(errors)
 return {'errors':errors,'MAE':Fraction(sum(abs(e) for e in errors),n),'MSE':Fraction(sum(e*e
for e in errors),n)}
# Added constructed fixture: original report supplies no numerical residuals.
fixture=error_metrics([1,3],[1,1])
assert fixture['errors']==[0,2] and fixture['MAE']==1 and fixture['MSE']==2
picture=
{'original_correct':5,'rerun_correct':4,'total':6,'disputed_case':'Q3','saved_file_available':'not
verified by this program'}
assert Fraction(picture['original_correct']-
picture['rerun_correct'],picture['total'])==Fraction(1,6)
print('Separate constructed metric fixture:',fixture)
print('Supplied six-case scoring disagreement:',picture)
print('Inspect saved Q3 output, reference label and rubric before adjudication. Do not choose
the larger score or average incompatible scoring decisions.')
print('If the saved Q3 file is missing, exact verification of that case is unresolved. Record
the missing artifact and any assistance; do not fabricate the file or a successful independent
check.')
```

```
print('No other operator, new model output, full-study reproduction or new-data replication is executed here. This local fixture only demonstrates MAE versus MSE.')
```

Evidence record and reflection

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

Predict, investigate and preserve the evidence

Plan → try → check

MAIN LESSON · SECTION 14

Try the selected new case independently

> Diagnose the changed validation split, identify the missing exact membership records and preserve both attempts without claiming an exact rerun.

Complete independent task

Reproducer uses different validation split because no split IDs were provided. Diagnose.

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Apply a new case independently

Attempt this without hints

Reproducer uses different validation split because no split IDs were provided. Diagnose.

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

MAIN LESSON · SECTION 11

Save the research artifact and unresolved conditions

> Keep supplied evidence, proposed revisions and actual work distinct.

Project task and artifact

For Reproducible AI Research Study, use this week's independent task as a rehearsal for the final **evidence** pack: Reproducer uses different validation split because no split IDs were provided. Diagnose.

Keep actual **observation**s separate from expected results, preserve failures, compare a baseline and explain the limits.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Evidence and limitations

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

▼ Earlier wording, alternatives and picture transcripts

Evidence for your project

MAIN LESSON · SECTION 16

Retrieve the mechanism and state the evidence boundary

> Explain what follows and what would require new evidence.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of independent reproduction check

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Use research vocabulary precisely**

REFERENCE · SECTION 5

Reference · Use research vocabulary precisely

> Connect each term to the question, comparison, measurement or evidence.

Vocabulary and your explanation

Word	Meaning
Independent operator	A person executing from the supplied record
Reproduction discrepancy	A mismatch needing documented investigation
Artifact manifest	A versioned inventory for the reported computation

Use two terms to explain a step in this week's example.

▼ **Extra challenge · Explain a stronger case**

REFERENCE · SECTION 10

Extra challenge · Explain a stronger case

> Choose the next challenge after the main route.

Explain,repair and extend

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

- A report omits this qualification: A successful reproduction repeats the original **evidence** and may preserve its original design limitations.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

Explain → apply → challenge

> **Explain which evidence supports your conclusion and which claim remains unproved.**

▼ **Reference · Check the research reasoning and its limits**

Reference · Check the research reasoning and its limits

> Use after practice; reserve the selected independent case for its first attempt.

Current explanation and response

- Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

  For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

> **Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.**

> **My limited claim and a useful next check**

REFERENCE · SECTION 13

Extra practice · Revisit the complete trace

> Predict before reading the supplied steps.

Complete trace and evidence

Complete evidence record

Independent reproduction check — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Operator gets different MSE because the supplied script averages absolute errors while paper names squared error.	Provide manifest, fixture data, code, environment and expected metric definition. Record assistance and mismatches; reproduction is not new-data replication.	Metric contract mismatch explains a candidate discrepancy. Correct/report script and rerun; preserve original artifact versions and results.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

▼ **Reference · The earlier route checklist**

REFERENCE · SECTION 15

Reference · The earlier route checklist

> Use one main route rather than repeating the lesson.

Targets,route and first prediction

Start with this question: An independent rerun with a disagreement. A second team follows the saved protocol on the same fixed data and obtains a different score.

By the end, I can	How I will show it
Explain the mechanism and terminology of independent reproduction check	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ **Extra practice · Predict,check and repair**

REFERENCE · SECTION 17

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- An independent rerun with a disagreement. A second team follows the saved protocol on the same fixed data and obtains a different score.
- Which labelled step matches this detail? Inspect Q3′s saved answer and rule; document whichever scoring decision is justified.
- 1. Resolve by the rubric
- 2. Compare exact artefacts
- 3. Locate disagreement
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Resolve by the rubric Inspect Q3's saved answer and rule; document whichever scoring decision is justified.

An independent reproduction check can uncover interpretation or implementation differences. Do not average incompatible scores or simply choose the higher one.

Changed-case question and optional hint

- Changed-case practice: Q3's saved file is missing from the archive. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. Mark exact reproduction incomplete until the required artefact is recovered or the limitation disclosed.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ Check after the changed attempt

After your attempt, compare with the supported explanation: Mark exact reproduction incomplete until the required artefact is recovered or the limitation disclosed. A missing record prevents checking the contested decision.

Repair and retain the record

Review Peer review and response (ai-7/32). Goal: Explain the **mechanism** and terminology of peer review and response. Short practice: Count photos and underlying plants in the reviewer's concern.

What analysis assumption needs revision, and why does changing only the word photos to plants fail to repair it?. Return to this week and explain what you changed.

Review Writing the methods section (ai-7/30). Goal: Explain the **mechanism** and terminology of writing the methods section. Short practice: Apply the inclusive four-character rule to oak, pine and willow.

Which label changes under greater-than-four, and what method detail allows another team to reproduce the distinction?. Return to this week and explain what you changed.

Review Robustness across conditions (ai-7/27). Goal: Explain the **mechanism** and terminology of robustness across conditions. Short practice: Calculate indoor, outdoor and combined rates.

Which relevant weakness disappears from view when only the combined rate is reported?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

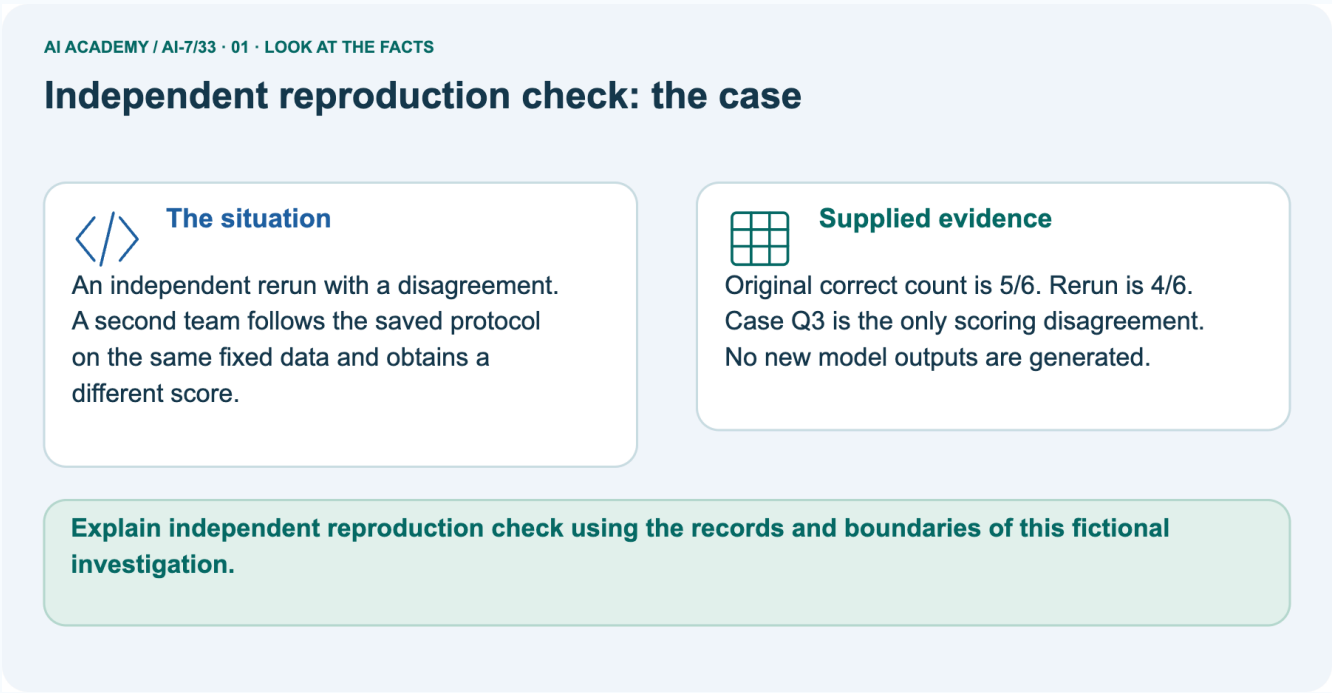
Keep your completed work

REFERENCE · SECTION 18

Reference · Six-picture concept journey

> Choose the picture that supports the current explanation.

01. Independent reproduction check: the case



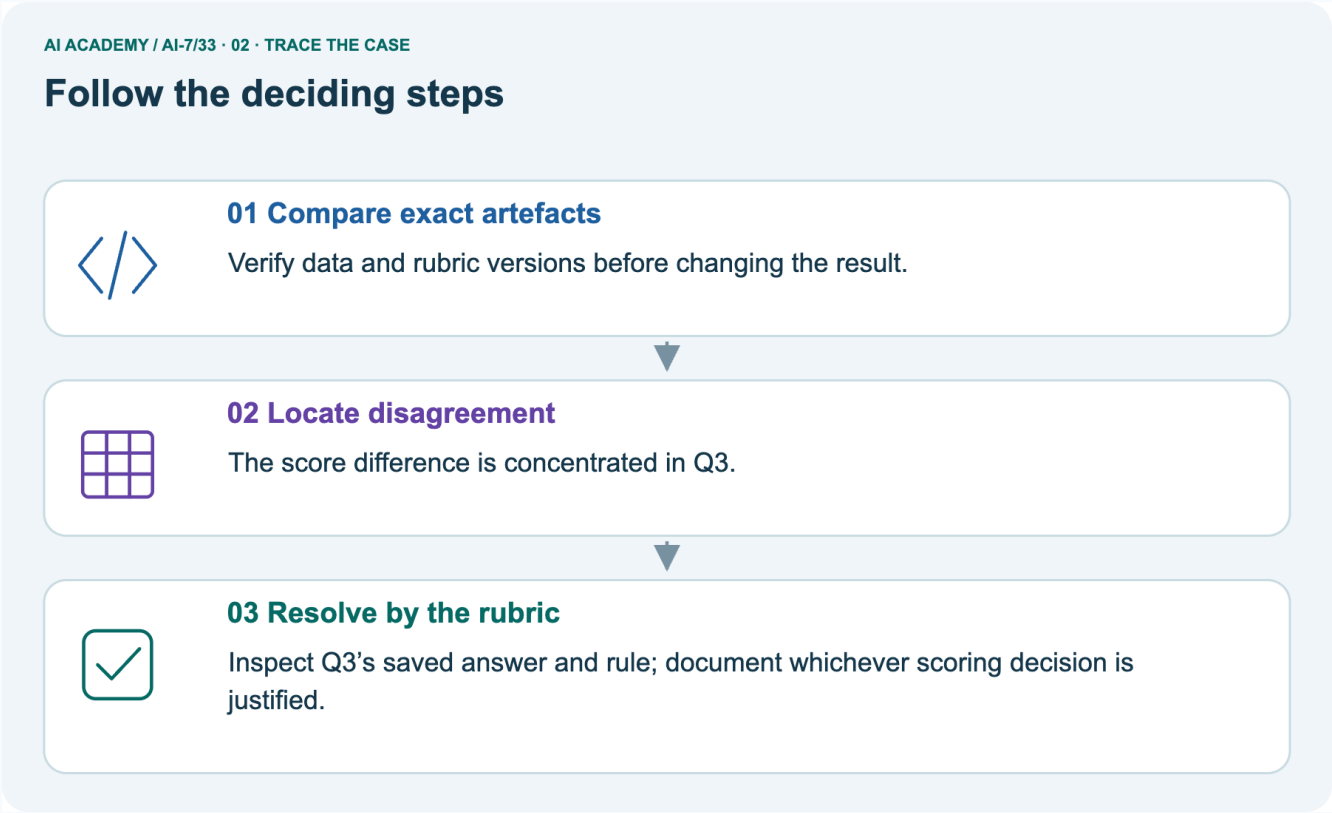
Follow the labelled picture with the matching explanation. Open the vector diagram

An independent rerun with a disagreement. A second team follows the saved protocol on the same fixed data and obtains a different score. Supplied facts: Original correct count is 5/6. Rerun is 4/6.

Case Q3 is the only scoring disagreement. No new model **output**s are generated.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

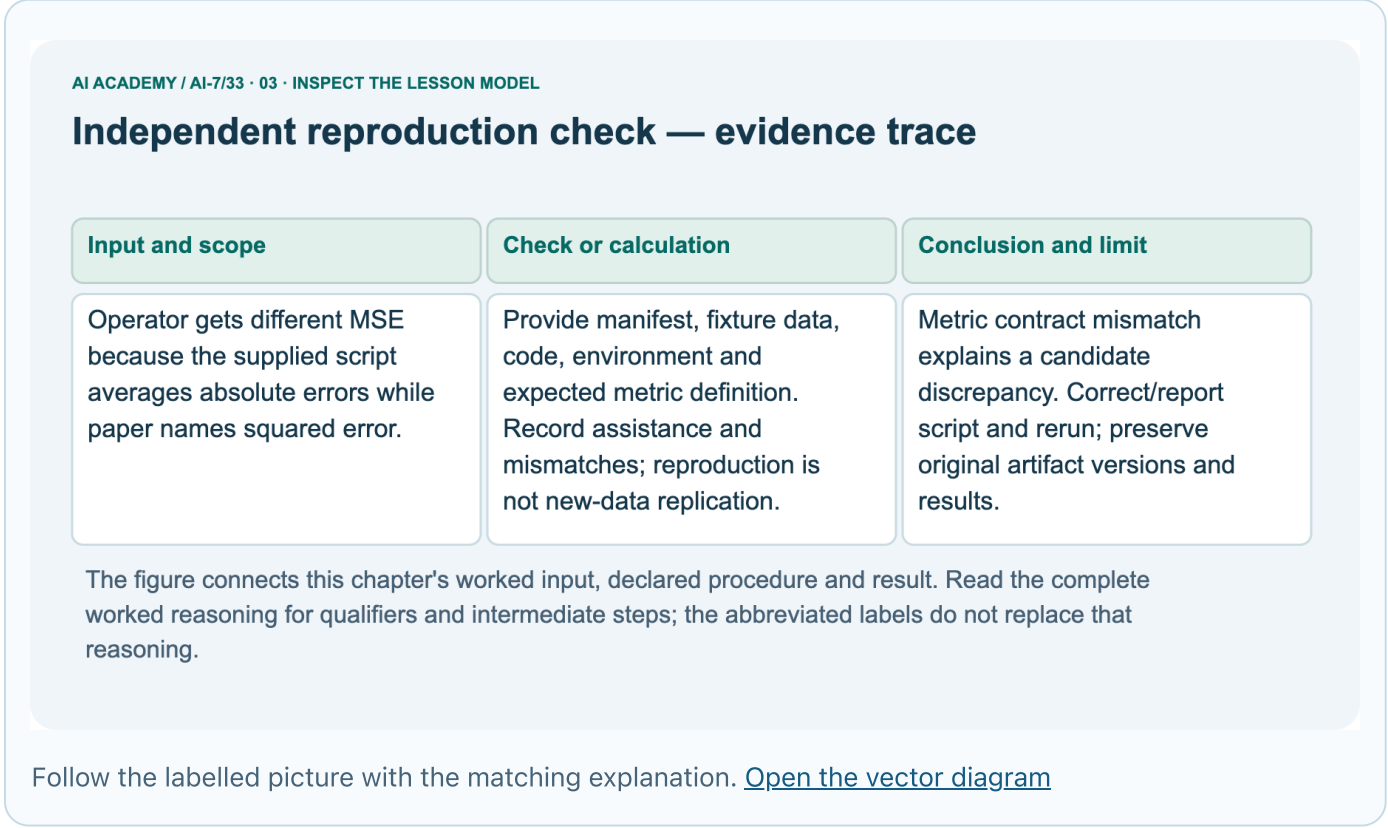


Follow the labelled picture with the matching explanation. Open the vector diagram

Compare exact artefacts: Verify data and rubric versions before changing the result. → Locate disagreement: The score difference is concentrated in Q3. → Resolve by the rubric: Inspect Q3's saved answer and rule; document whichever scoring decision is justified.

Point to each numbered stage and explain which recorded **input** it uses.

03. Independent reproduction check — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

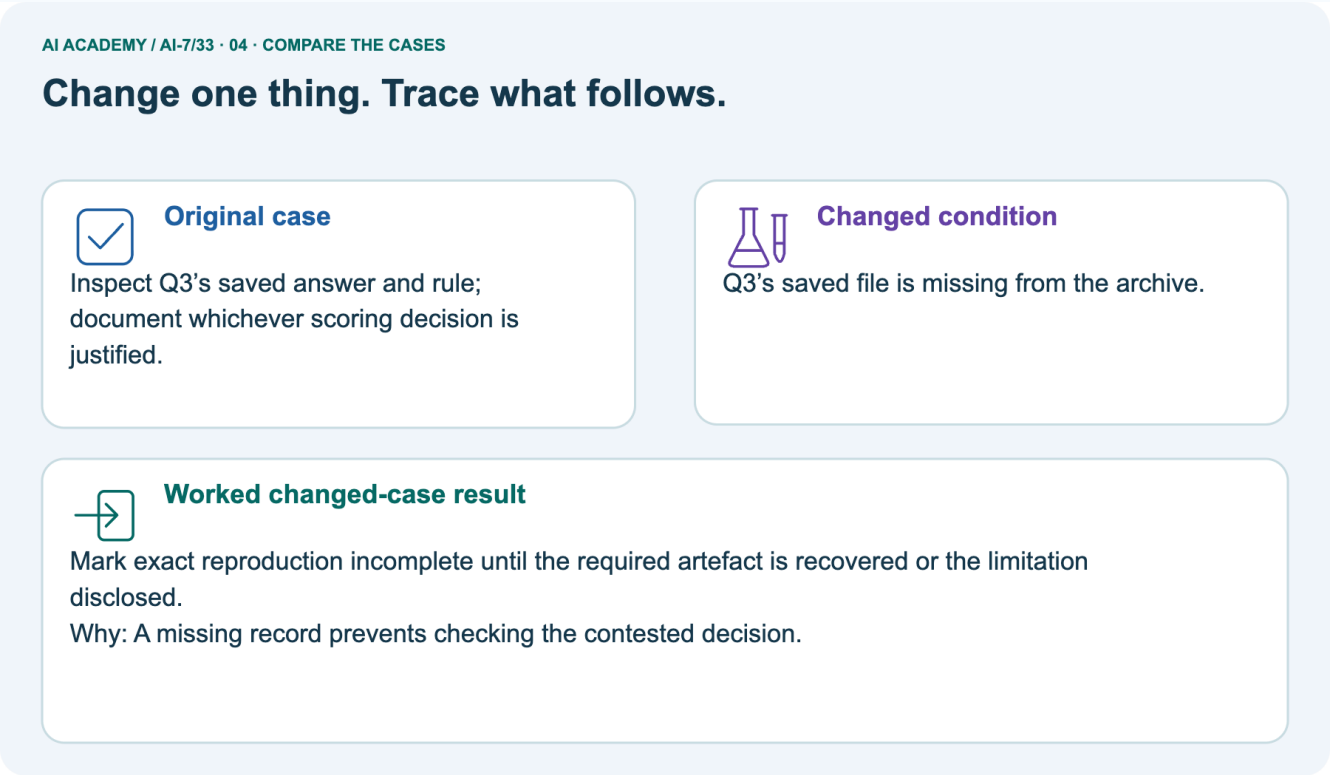
Independent reproduction check — **evidence** trace.

Operator gets different MSE because the supplied script averages absolute errors while paper names squared error. | Provide manifest, fixture data, code, environment and expected metric definition.

Record assistance and mismatches; reproduction is not new-data replication. | Metric contract mismatch explains a candidate discrepancy. Correct/report script and rerun; preserve original artifact versions and results.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



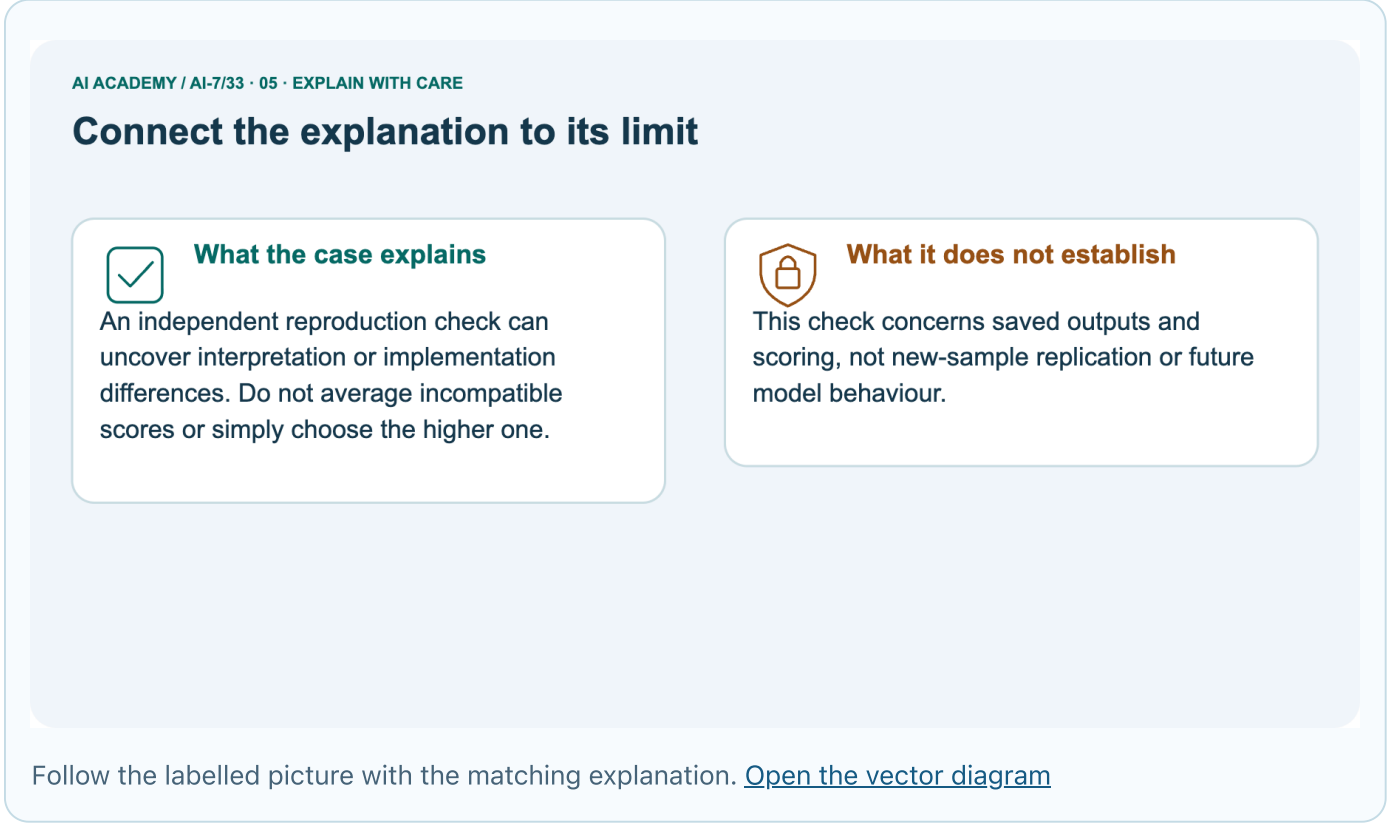
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Inspect Q3′s saved answer and rule; document whichever scoring decision is justified. Changed condition: Q3′s saved file is missing from the archive.

Expected result: Mark exact reproduction incomplete until the required artefact is recovered or the limitation disclosed. Reason: A missing record prevents checking the contested decision.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



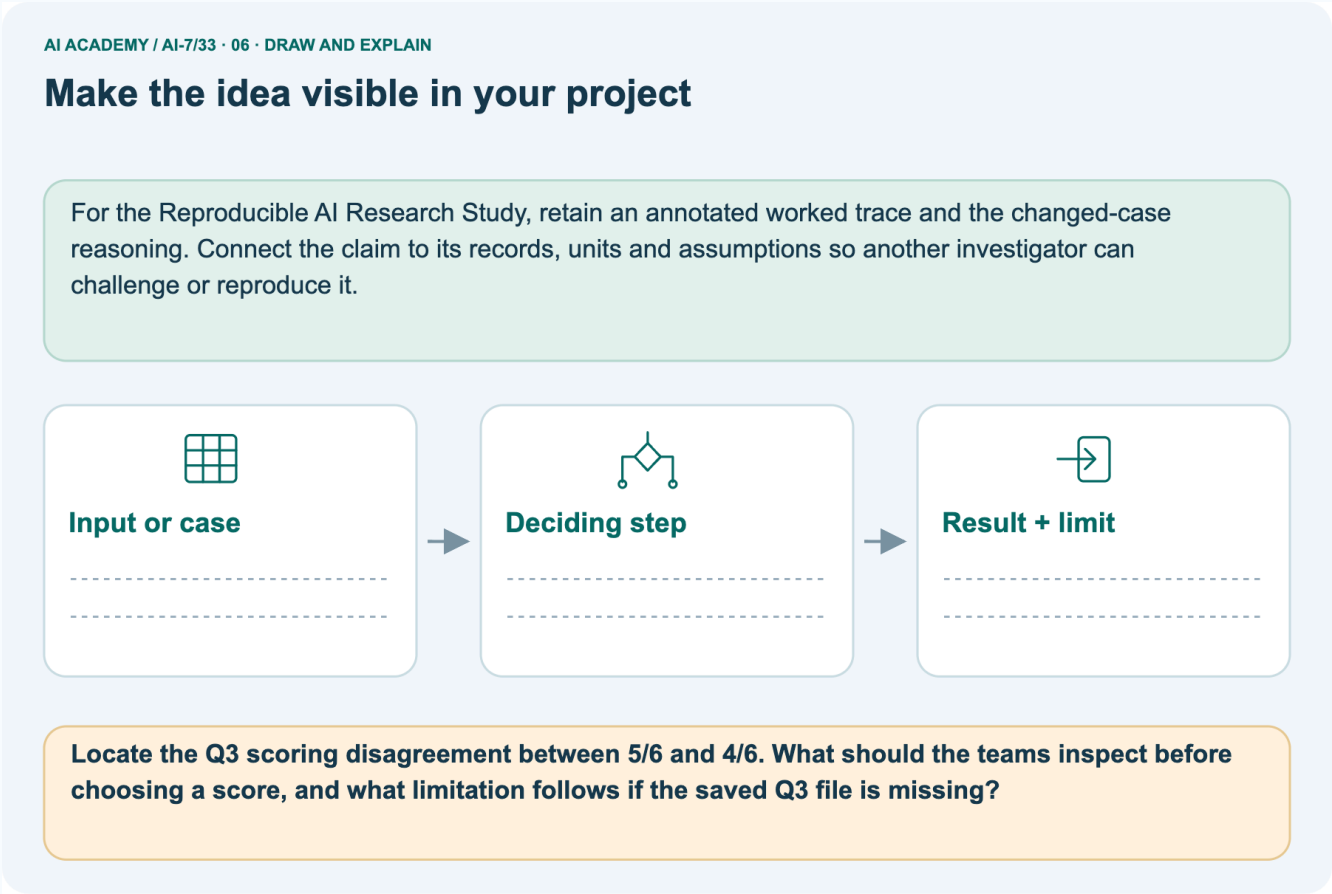
Follow the labelled picture with the matching explanation. Open the vector diagram

An independent reproduction check can uncover interpretation or implementation differences. Do not average incompatible scores or simply choose the higher one.

Limit: This check concerns saved **output**s and scoring, not new-sample replication or future model behaviour.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Locate the Q3 scoring disagreement between 5/6 and 4/6.

What should the teams inspect before choosing a score, and what limitation follows if the saved Q3 file is missing?

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Independent reproduction check: the case

02. Follow the deciding steps

03. Independent reproduction check — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.